LESSON PLAN

EJECTION SEAT (36)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Definitions of Atomic structure, Nucleus Protons, Neutrons Electrons Atomic mass Isotopes Mass defect Nuclear fission Nuclear fusion Fission
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Definitions of Atomic structure, Nucleus Protons, Neutrons Electrons Atomic mass Isotopes Mass defect Nuclear fission Nuclear fusion Fission –fusion –fission
5. Specific Objectives	36.1.1.1 Trainee will be able to explain conventional and nuclear explosion

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of atomic diagrams; Instructor supervision required for safety concepts
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the basic unit of matter?
(b) Define the term 'isotope'.
(c) What is the primary difference between conventional and nuclear weapons?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Atomic structure definitions • Nucleus components protons neutrons • Electrons atomic mass concepts • Isotopes and mass defect explanation	L & Ds	PPT, IB	15 min
(b)	• Nuclear fission process definition • Nuclear fusion process definition • Mechanism of energy release • Comparison of fission and fusion	L & Ds	PPT, PDF	20 min
(c)	• Conventional vs nuclear explosion • Chain reaction principles • Weapon yield factors • Safety implications of nuclear material	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between mass defect and atomic mass.
(b) Describe how a chain reaction occurs in nuclear fission.
(c) Differentiate between the energy sources in conventional and nuclear explosions.

12. Home Assignment and Follow Up	Prepare a brief note on the basic requirements of a fission weapon for the next lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Basic requirements of a fission weapons. Critical mass of fissile material. Basic working principles of fission weapons	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min